

Weekly Lesson Plan – Week 9 · Semester 1 · 2026–27 · Da Vinci School

Teacher: Gavaldon Week of: October 5, 2026 Course: Engineering Design & Problem Solving (InSPIRESS)				
TEKS / ELPS:				
TEKS: §127.406(6) implement a design process to develop a multi-system product or solution; 127.406(4) communicate findings using appropriate tools ELPS: c.1C, c.2C, c.3E, c.4C, c.4F, c.5G				
Aim of the Lesson / Objective: <i>Statement of what you want students to do (skill) and/or know (knowledge). Students will...</i>				
MONDAY OCT 5	TUESDAY OCT 6	WEDNESDAY OCT 7	THURSDAY OCT 8	FRIDAY OCT 9
Students will complete the paperclip model by sketching a fully defined path and sweeping a profile along it. ----- ALTERNATE: Students will write 3-4 paragraphs analyzing how automation and artificial intelligence change employment, and post photographic evidence of the day's work.	Students will create sketch profiles on offset construction planes and join them into one body with the loft feature. ----- ALTERNATE: Students will write 3-4 paragraphs analyzing how automation and artificial intelligence change employment, and post photographic evidence of the day's work.	Students will refine a lofted body with guide rails created from an intersect, then shell it and add modeled threads. ----- ALTERNATE: Students will write 3-4 paragraphs analyzing how automation and artificial intelligence change employment, and post photographic evidence of the day's work.	Students will model the base of an ice cube tray using the block method, a tapered extrude, and dimensions linked with fx. ----- ALTERNATE: Students will write 3-4 paragraphs analyzing how automation and artificial intelligence change employment, and post photographic evidence of the day's work.	Students will pattern a feature with the correct compute type and finish a thin-walled part using the shell command. ----- ALTERNATE: Students will write 3-4 paragraphs analyzing how automation and artificial intelligence change employment, and post photographic evidence of the day's work.
Employed Language Domain(s) Objective: <i>How students achieve the objective – Listening, Speaking, Reading, Writing. Students will...</i>				
Students will collaborate in teams to investigate engineering problems, communicate their findings, and present their solutions. They will listen to team members' ideas, discuss potential solutions, read relevant technical materials, and write explanations of their designs and findings. Alternate-track students read, write and revise a multi-paragraph study of automation and employment, and present their daily work in writing on Google Classroom.				
Assessment of Content Objective: <i>Instrument used to determine mastery. Ex: Exit Ticket.</i>				
Main project: the model file in each student's own cloud folder, checked against the day's feature list and the timeline. Alternate assignment: a daily photo of 3-4 written paragraphs posted to Google Classroom, and a culminating conclusion on Friday.				
Set (Bell Work): <i>A motivating action that engages students in the objective – question, fact, activity. Essential Question:</i>				
MONDAY OCT 5	TUESDAY OCT 6	WEDNESDAY OCT 7	THURSDAY OCT 8	FRIDAY OCT 9
a paperclip has not changed in 100 years - why has nobody improved it?	why does a bottle change shape from square at the base to round at the neck?	a part passes testing but you know the wall is too thin - do you ship it?	why is every ice cube tray pocket wider at the top than at the bottom?	a factory buys one machine that replaces four people - what does the town lose?
Methodology / Steps: <i>What it takes to teach the objective so 100% can pass the assessment. Teacher will... Teams will... Individuals will...</i>				
MONDAY OCT 5	TUESDAY OCT 6	WEDNESDAY OCT 7	THURSDAY OCT 8	FRIDAY OCT 9
Step 1: Bell ringer / essential question. Step 2: Bell work: a paperclip has not changed in 100 years - why has nobody improved it? (10 min) Step 3: Day 3 video: finish the path, the circle profile on the XZ plane, then Sweep (28 min) Step 4: Fix-it round: sweep failures (open path, wrong plane, chain selection) (5 min) Step 5: Save (Ctrl+S) in Students > your name (2 min) ----- ALTERNATE ASSIGNMENT: Automation Study Day 1: what machines already replaced - 3 to 4 paragraphs, plus a photo of today's writing posted to Classroom.	Step 1: Bell ringer / essential question. Step 2: Bell work: why does a bottle change shape from square at the base to round at the neck? (10 min) Step 3: Offset construction planes: why a sketch does not have to sit on the origin (8 min) Step 4: Day 4 video: four profiles, then Loft (22 min) Step 5: Save (Ctrl+S) (5 min) ----- ALTERNATE ASSIGNMENT: Automation Study Day 2: the company's side - cost, speed and quality - 3 to 4 paragraphs + photo posted.	Step 1: Bell ringer / essential question. Step 2: Bell work (dilemma): a part passes testing but you know the wall is too thin - do you ship it? (10 min) Step 3: Day 4 video: Intersect for rails, mirror the rails, re-loft, shell, threads (25 min) Step 4: Fix-it round: loft errors and shell errors on curved bodies (8 min) Step 5: Save (Ctrl+S) - Day 4 file in your folder (2 min) ----- ALTERNATE ASSIGNMENT: Automation Study Day 3: the worker's side - who loses the job, what retraining really costs - 3 to 4 paragraphs + photo posted.	Step 1: Bell ringer / essential question. Step 2: Bell work: why is every ice cube tray pocket wider at the top than at the bottom? (10 min) Step 3: Day 5 video: center rectangle, extrude the base, first cutout, taper, fx dimensions (25 min) Step 4: Internal fillets; check your taper direction (8 min) Step 5: Save (Ctrl+S) (2 min) ----- ALTERNATE ASSIGNMENT: Automation Study Day 4: AI now - what is different this time, and which jobs are actually exposed - 3 to 4 paragraphs + photo posted.	Step 1: Bell ringer / essential question. Step 2: Bell work: a factory buys one machine that replaces four people - what does the town lose? (10 min) Step 3: Day 5 video: rectangular pattern (compute types), then the shell (22 min) Step 4: Folder check: Day 3, Day 4 and Day 5 files all in Students > your name (8 min) Step 5: Three weeks of tools: extrude, revolve, sweep, loft, shell - one line each (5 min) ----- ALTERNATE ASSIGNMENT: Automation Study Day 5: culminating conclusion, 3 paragraphs, plus all 5 photos turned in on Classroom.
Resources: <i>Textbook, handouts, visuals, laptops.</i>				
Google Classroom, projector, laptops, Autodesk Fusion (desktop or fusion.online.autodesk.com), the "Learn Autodesk Fusion in 30 Days" 2026 Edition videos (Days 3, 4, 5), engineering notebook, the alternate-assignment prompt sheet.				
Re-Teaching Activities: <i>What you prepare if students do not master the objective.</i>				
Rewind-and-rebuild at the teacher table for the feature that failed; a printed step list for the same build; one-on-one at the student's screen; peer re-teach inside the seating group. Alternate track: sentence stems, a word bank, and a one-on-one conference on the day's prompt.				
Reflection: <i>Students analyze what they accomplished and how they reached mastery.</i>				
Exit ticket – students state which feature they finished and the one mistake they fixed. Alternate track: students state the claim they argued that day and the evidence they used.				
Study Skills: <i>Skills implemented by students – note-taking, summarizing, teamwork, etc.</i>				
Note-taking, summarizing, following multi-step technical directions, self-checking against a list, teamwork, written argument.				
SPED & 504 Accommodations / Modifications: <i>Codes.</i>				
Per Individualized Education Plan (IEP) / 504 plan.				
ELL Accommodations: <i>Codes.</i>				
Clarify directions (CD); Extra time (ET); Peer and Native Language Support (PN); Pre-teach vocabulary (PV); Rephrase, Repeat, Slow Down (RS); Visual Cues (VC); Wait time (WT); Pictorial Representation (DP). GROUPING: students are seated in mixed-proficiency small groups so emergent bilingual students hear and use the academic vocabulary in context, rehearse with a peer before whole-class response, and can be pulled as a small group for re-teach (PN, WT). ONE-ON-ONE: individual teacher explanation is available on request every period; directions are restated and modeled individually at the student desk after whole-class delivery (CD, RS, VC).				